



RUB

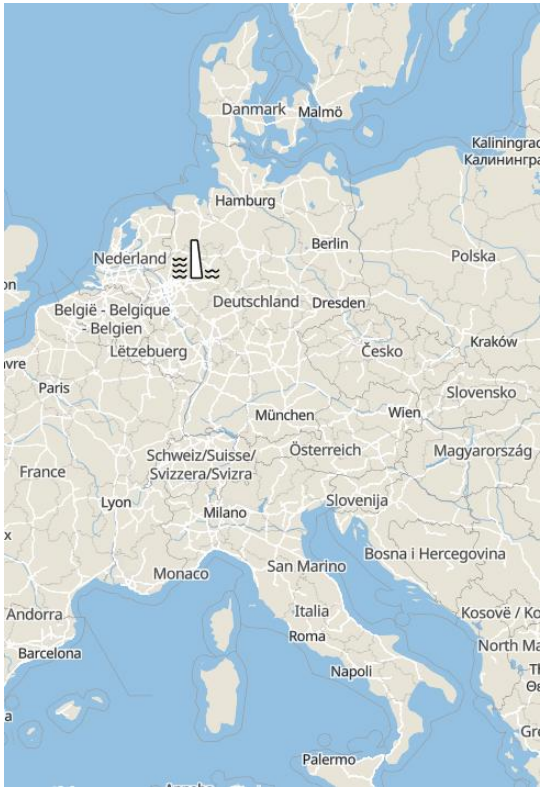
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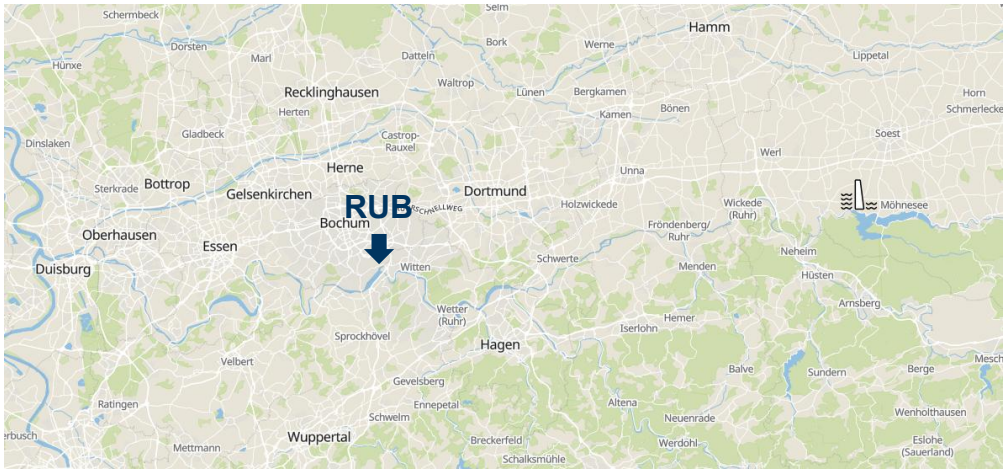
WORLDQUAL LIGHT EXCERCISE

TP Loadings in the Möhne Lake Catchment

Katrin Schweden

The Möhne chatchment and reservoir





Size catchment: 436,29 km²

Reservoir „Volume“: 134,5 Mio m²

What WorldQual light computes

Total phosphorus (TP) loads per grid cell and month, aggregated to the basin

Eight input pathways: three point sources, five diffuse sources

Grid: 5 arcmin (~10 × 10 km), 180,721 model cells across Europe

Runs fully offline from the bundled data package, no database and no VPN needed

Python requirements

What gets installed

- pandas, numpy, scipy, pyyaml
- matplotlib, seaborn (plots)
- geopandas, pyogrio (GeoPackage)
- mysql-connector-python, python-dotenv

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Data files

```
WorldQualLite/
├── config.yaml                    all settings for one model run
├── run.py                         the runner
├── src/                           the six model modules
└── data/
    ├── lookups/                   offline lookup tables (no database needed)
    ├── output/                    results land here
    └── sample_run/
        ├── basin/basin_cells.csv  the cells of the basin
        ├── cell_input/            population, return flows per cell
        └── unf/                   binary raster data (runoff, livestock, ...)
```


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Setup and configuration in config.yaml

```yaml
run:
  type: "Historical"
  data_source: "Excel"

time:
  initial_year: 2011
  final_year: 2011

spatial:
  country_id: 250
  continent_index: 0
  parameter_id: 60
  downstream_cells: [132488]
```

Start a model run

Start: run.py with config.yaml

What happens, year by year

- Delineate the basin from downstream_cells if basin_cells.csv is missing
- Load the UNF rasters: runoff, urban runoff, livestock, fertilizer, cropland
- Fetch country and cell data: population, treatment rates, emission factors
- Compute the eight load sources per cell and month
- Apply the erosion factor (Fink et al.) and retention (Vollenweider)
- Aggregate to the basin and write CSV, GeoPackage and plot

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Results

Everything lands in the configured output_folder, here data/output_rhone/.

File	Content
TP_yearly_results.csv	one row per year, one column per input pathway, basin total
TP_cell_results.csv	one row per cell and year, plus GCRC_EU as the QGIS join key
TP_cell_results.gpkg	the same cells with geometry, only when paths.grid_gpkg is set
TP_yearly_plot.png	stacked bar chart of the annual loads
TP_validation_plot.png	measured versus simulated, only with measurement data

Units: t/year, t/month per cell

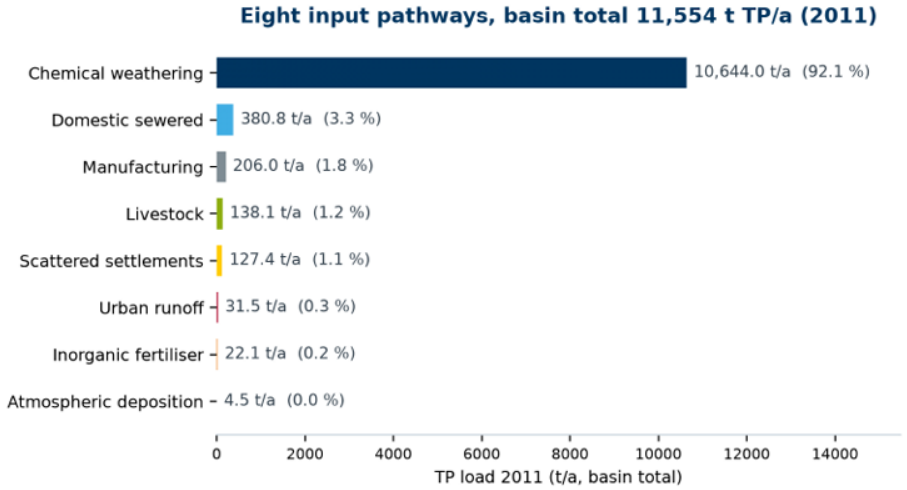
```
Year,DomesticSeweredLoad,ManufacturingLoad,Scattered Load,InorganicFertilizerLoad,
AgricultureLivestockLoad,AtmBackgroundLoad,CWBackgroundLoad,UrbanSurfaceRunoffLoad
2011,380.78,205.98,127.40,22.08,138.10,4.49,10643.97,31.49
```

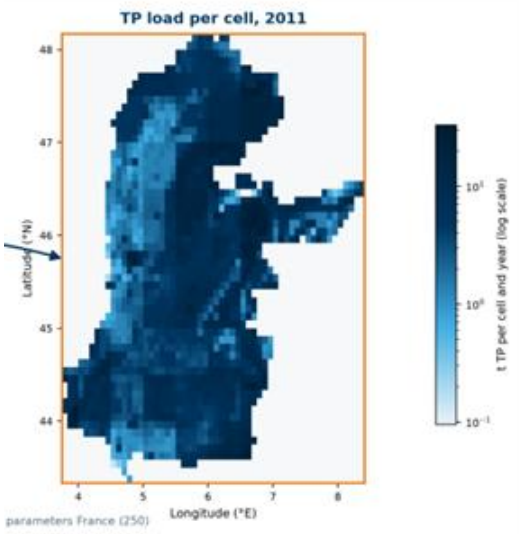
How to Work with the Output

```
import geopandas as gpd, pandas as pd
grid  = gpd.read_file("data/grid_eu/grid_eu.gpkg", layer="zellgitter_gcrc")
cells = pd.read_csv("data/output_rhone/TP_cell_results.csv")
gdf   = grid.merge(cells, on="GCRC_EU")      # never join on Cell_ID
gdf.to_file("rhone_tp_2011.gpkg", driver="GPKG")
```

Join on GCRC_EU, never on Cell_ID. A join on the wrong key gives a map that looks plausible and sits hundreds of kilometres off.

Examples for Presentation of the Results





- Gridded loadings per year, three year median in total, or for every single load type
- Loadings per year as a timeseries in total or every single load

Validation of Wastewater Loadings



Wastewater Treatment Plants

Total Phosphor Loadigns = Water volume x Concentration

!WorldQual gives you monthly loadings!
!Check for differences in units!

How to validate?

1. Calculate loadings for measured data
2. Identify cells with wastewater treatment plants
3. Compare calculated loads for wasterwater with modelled loads

Scenarios summary

Groups:

Group 1: SSP1 and RCP 2.6

Group 2: SSP5 and RPC 8.5

What to do with the results?

- **Identify trends and changes in different loading sources**
- **How is the spacial distribution of the loadings in comparison to the historic dataset. Compare median of 2000/2001/2002, 2029/2030/2031 and 2048/2049/2050 by median of 36 month.**
- **Compare your results with the other group**

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